

화학수학

열네 번째 수업

연산자

Mathematical Operators (연산자)

perform **mathematical operations** on some function

e.g. "+3" operator, "×2" operator, "d/dx" operator

- Typically denoted by a letter with a **caret** (^)
- Operate on the function on its right

Special Operators

- Identity operator (항등 연산자) $\hat{E}$: multiplication by 1 (doing nothing)

$$\hat{E}f = f$$

- Null operator (영 연산자) $\hat{0}$: multiplication by 0

$$\hat{0}f = 0$$

Operator Algebra

- Sum of two operators: if $\hat{C} = \hat{A} + \hat{B}$, then

$$\hat{C}f = (\hat{A} + \hat{B})f = \hat{A}f + \hat{B}f$$

- Product of two operators: if $\hat{C} = \hat{A}\hat{B}$, then

$$\hat{C}f = \hat{A}(\hat{B}f)$$

Properties of Operator Multiplication

- Associative property (결합법칙)

$$(\hat{A}\hat{B})\hat{C} = \hat{A}(\hat{B}\hat{C})$$

- Distributive property (분배법칙)

$$\hat{A}(\hat{B} + \hat{C}) = \hat{A}\hat{B} + \hat{A}\hat{C}$$

- Commutative property (교환법칙) → Not necessarily

$$\hat{A}\hat{B} = \hat{B}\hat{A} ?$$

More on Commutative Property

- If $\hat{A}\hat{B} = \hat{B}\hat{A}$, $\hat{A}$ and $\hat{B}$ are said to **commute**
- **Commutator** (교환자)

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$$

- If $\hat{A}$ and $\hat{B}$ commute,

$$[\hat{A}, \hat{B}] = \hat{0}$$

Exponentiation of Operator

(연산자의 거듭제곱)

$$\hat{A}^n = \hat{A}\hat{A} \cdots \hat{A} \text{ (} n \text{ times)}$$

Inverse of Operator (역 연산자)

- $\hat{A}^{-1}$: inverse of operator $\hat{A}$

$$\hat{A}\hat{A}^{-1} = \hat{A}^{-1}\hat{A} = \hat{E}$$

- $\hat{A}$ is also the inverse of $\hat{A}^{-1}$
- Not all operators have inverses (*e.g.* $\hat{0}$)

Operator Algebra and Differential Equation

- Derivative operator $\hat{D}_x = d/dx$
- Homogeneous linear DE with constant coefficients

$$a_n \frac{d^n y}{dx^n} + a_{n-1} \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_1 \frac{dy}{dx} + a_0 y = 0$$

$$(a_n \hat{D}_x^n + a_{n-1} \hat{D}_x^{n-1} + \cdots + a_1 \hat{D}_x + a_0 \hat{E})y = 0$$

This can be solved by operator algebra

Eigenfunctions and Eigenvalues

(고유함수와 고윳값)

For a given operator $\hat{A}$, a function f such that

$$\hat{A}f = af$$

is called an **eigenfunction** of $\hat{A}$, and a is called an **eigenvalue**. This equation is called an **eigenvalue equation** (고윳값 방정식).

Operators in Quantum Mechanics

“For every **observable quantity**,
there is a corresponding **mathematical operator**”

- Position operators: $\hat{x} = x$, $\hat{y} = y$, $\hat{z} = z$
- Momentum operators:

$$\hat{p}_x = \frac{\hbar}{i} \frac{\partial}{\partial x}, \quad \hat{p}_y = \frac{\hbar}{i} \frac{\partial}{\partial y}, \quad \hat{p}_z = \frac{\hbar}{i} \frac{\partial}{\partial z}$$

- System energy operator (Hamiltonian):

$$\hat{H} = -\frac{\hbar^2}{2m} \nabla^2 + \mathcal{V}(x, y, z)$$

Hermitian Property (에르미트 성질)

- If $\hat{A}$ is **Hermitian**, it obeys the relation

$$\int \chi^* (\hat{A}\psi) dq = \int (\hat{A}\chi)^* \psi dq$$

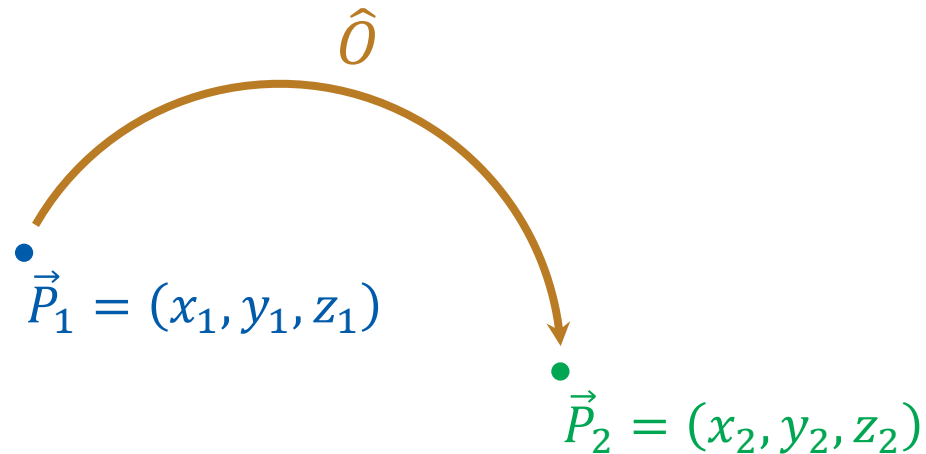
where χ and ψ are arbitrary functions and q represents all of the coordinates.

- All of the quantum-mechanical operators are Hermitian

오늘의 수업

- Symmetry operators
 - Point symmetry operators
 - Symmetry operators on functions
 - Eigenfunctions of symmetry operators

Symmetry Operators



$$\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\hat{O}\vec{P}_1 = \vec{P}_2$$

Point Symmetry Operators

(점대칭 연산자)

a class of symmetry operators
that leave **at least one point unmoved**
(the unmoved point = origin)

Identity operator

Inversion operator

Reflection operator

Rotation operator

Identity Operator and Inversion Operator

(항등 연산자 및 반전 연산자)

- Identity operator $\hat{E}$: doing nothing

$$\hat{E}(x_1, y_1, z_1) = (x_1, y_1, z_1)$$

$$\hat{E}\vec{r}_1 = \vec{r}_1$$

- Inversion operator $\hat{i}$: inversion w.r.t. the origin

$$\hat{i}(x_1, y_1, z_1) = (-x_1, -y_1, -z_1)$$

$$\hat{i}\vec{r}_1 = -\vec{r}_1$$

Reflection Operators (반사 연산자) $\hat{\sigma}$

reflection through a plane

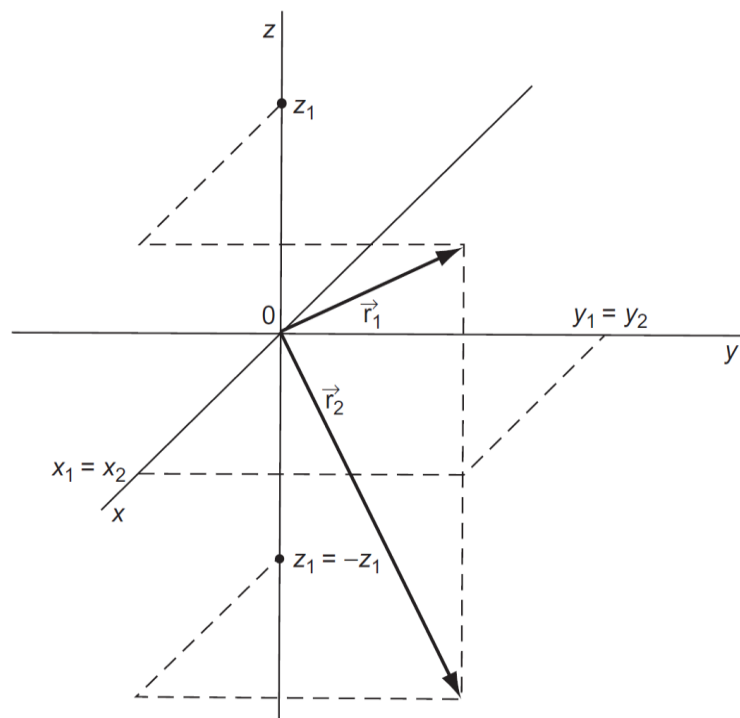
(The plane must be specified)

$$\hat{\sigma}_{(xy)}(x_1, y_1, z_1) = (x_1, y_1, -z_1)$$

$$\hat{\sigma}_{(xz)}(x_1, y_1, z_1) = (x_1, -y_1, z_1)$$

$$\hat{\sigma}_{(yz)}(x_1, y_1, z_1) = (-x_1, y_1, z_1)$$

Action of $\hat{\sigma}_{(xy)}$

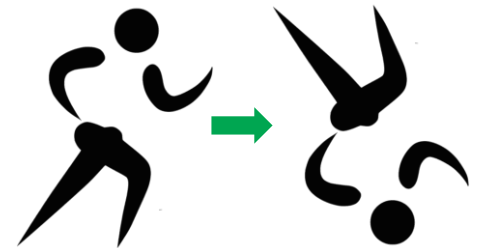


(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.1)

Horizontal and Vertical Reflections

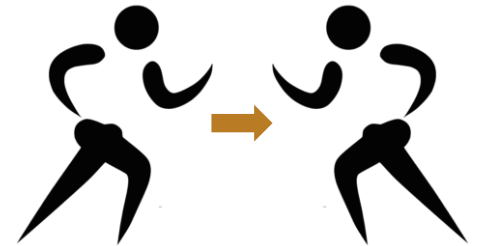
If the system has a principal axis,
we normally regard it as **the z axis**

- Horizontal reflection $\hat{\sigma}_h = \hat{\sigma}_{(xy)}$ (unique)



- Vertical reflection $\hat{\sigma}_v$: other reflections

$$\hat{\sigma}_{v(xz)}, \hat{\sigma}_{v(yz)}, \dots$$



Proper Rotation Operators $\hat{C}_n$

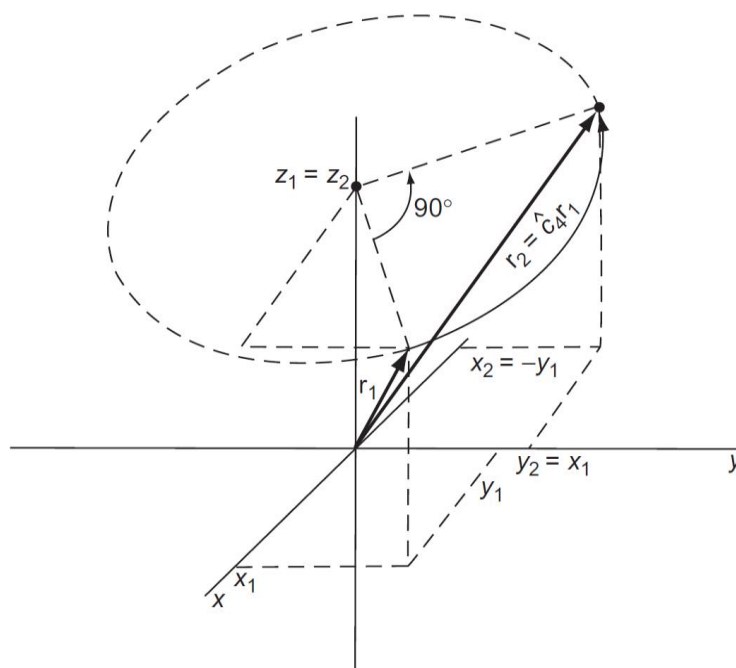
(단순 회전 연산자)

Counterclockwise rotation by $2\pi/n$

- n applications of the rotation operator = 1 full rotation
- The rotation axis must be specified

$$\hat{C}_{4(z)}(x_1, y_1, z_1) = (-y_1, x_1, z_1)$$

Action of $\hat{C}_{4(z)}$



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.2)

Improper Rotation Operators $\hat{S}_n$

(반사 회전 연산자)

$\hat{C}_n$ followed by $\hat{\sigma}$

(through a plane perpendicular to the rotation axis)

- The rotation axis must be specified

$$\hat{S}_{4(z)}(x_1, y_1, z_1) = (-y_1, x_1, -z_1)$$

Examples 13.10 and 13.11

Find $\hat{C}_{4(z)}(1, -4, 6)$ and $\hat{S}_{4(z)}(1, 2, 3)$.

Since $\hat{C}_{4(z)}(x_1, y_1, z_1) = (-y_1, x_1, z_1)$,

$$\hat{C}_{4(z)}(1, -4, 6) = (4, 1, 6)$$

Since $\hat{S}_{4(z)}(x_1, y_1, z_1) = (-y_1, x_1, -z_1)$,

$$\hat{S}_{4(z)}(1, 2, 3) = (-2, 1, -3)$$

Symmetry Elements (대칭 요소)

Point, line, or plane relative to which the symmetry operation is performed

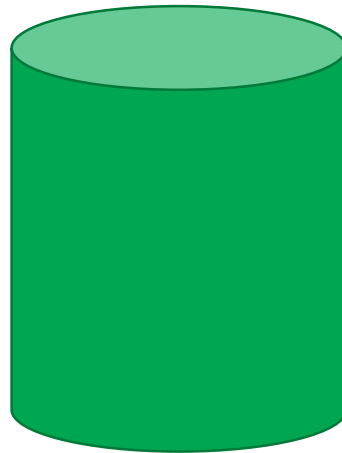
- Must include the origin
- Sometimes denoted by the same symbol
- Inversion i : the origin
- Reflection σ : the reflection plane
- Rotation C_n, S_n : the rotation axis

Symmetry of Object

- If after a symmetry operation, all particles in the object is in the same conformation as before, **the symmetry operator belongs to the object.**
- Coordinate system
 - origin: center of mass of the object
 - z axis: the rotation axis of highest order (largest n)

Example 13.12

List the symmetry operators belonging to a right circular cylinder.



$$\hat{I}, \hat{C}_{\infty(z)}, \hat{S}_{\infty(z)}, \hat{\sigma}_h, \infty \hat{\sigma}_v, \infty \hat{C}_2, \infty \hat{S}_2$$

Symbols for Rotational Symmetry



C_2 axis



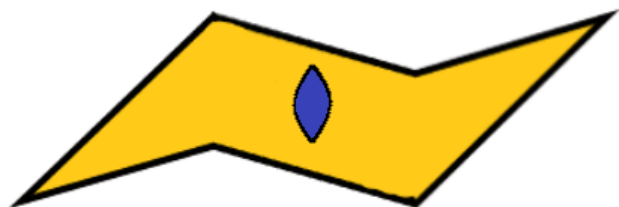
C_3 axis



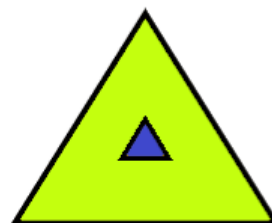
C_4 axis



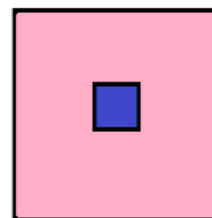
C_6 axis



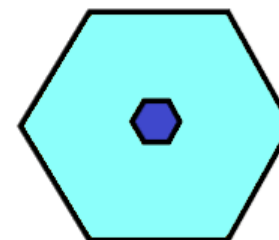
2-fold



3-fold



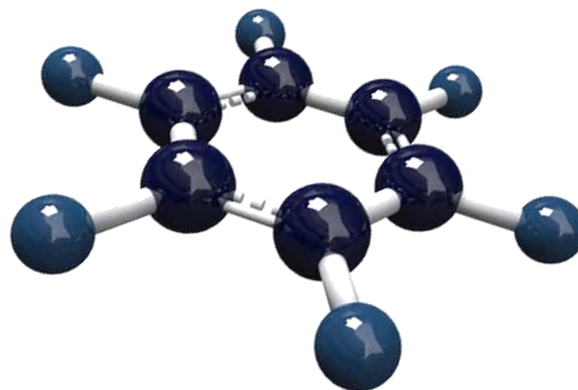
4-fold



6-fold

Example 13.13

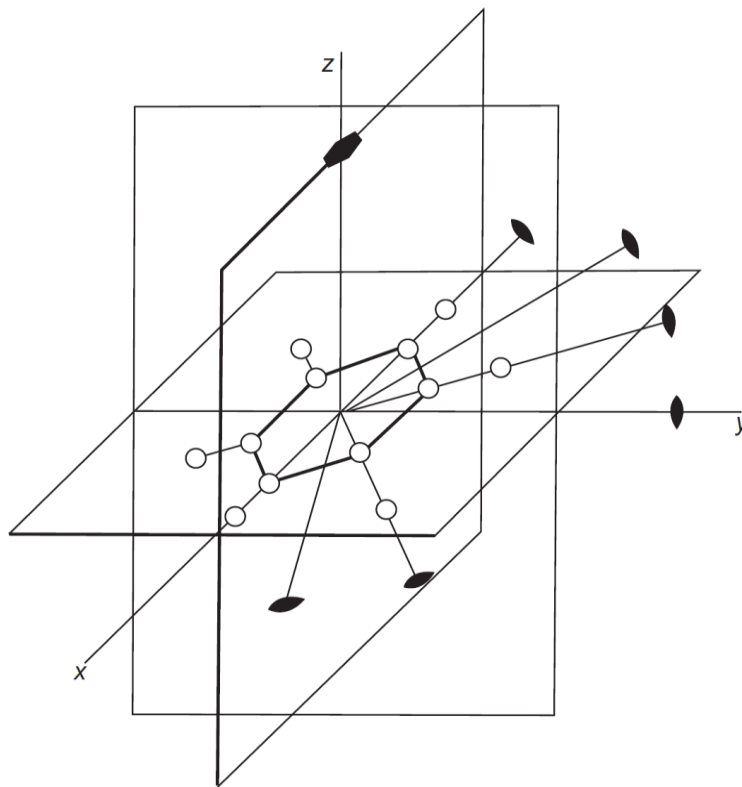
List the symmetry elements of the benzene molecule in its equilibrium conformation.



$$\hat{i}, \hat{C}_{6(z)}, \hat{S}_{6(z)}, \hat{\sigma}_h, 6\hat{\sigma}_v, 6\hat{C}_2, 6\hat{S}_2$$

Example 13.13

List the symmetry elements of the benzene molecule in its equilibrium conformation.



(Image: Mortimer, *Mathematics for Physical Chemistry*, 4th ed., Figure 13.3)

Symmetry Operators on Functions

If an operator $\hat{O}$ operates as $\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$,
for a function f ,

$$\hat{O}f = g$$

where $g(x_2, y_2, z_2) = f(x_1, y_1, z_1)$.

Use the relation between (x_1, y_1, z_1) and (x_2, y_2, z_2) !

Example 13.14

The normalized $2p_x$ wave function for the electron in a hydrogen atom is

$$\psi_{2p_x}(x, y, z) = x \exp \left[-\frac{(x^2 + y^2 + z^2)^{1/2}}{2a_0} \right],$$

where a_0 is the Bohr radius. Find $\hat{C}_{4(z)}\psi_{2p_x}$.

Since $\hat{C}_{4(z)}(x_1, y_1, z_1) = (x_2, y_2, z_2) = (-y_1, x_1, z_1)$, convert

$$x_2 \leftarrow -y_1, \quad y_2 \leftarrow x_1, \quad z_2 \leftarrow z_1;$$

$$\psi_{2p_x}(x_1, y_1, z_1) = x_1 \exp \left[-\frac{(x_1^2 + y_1^2 + z_1^2)^{1/2}}{2a_0} \right]$$

$$(\hat{C}_{4(z)}\psi_{2p_x})(x_2, y_2, z_2) = y_2 \exp \left[-\frac{(y_2^2 + (-x_2)^2 + z_2^2)^{1/2}}{2a_0} \right]$$

Example 13.14

The normalized $2p_x$ wave function for the electron in a hydrogen atom is

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$$(\hat{C}_{4(z)}\psi_{2p_x})(x_2, y_2, z_2) = y_2 \exp \left[-\frac{(y_2^2 + (-x_2)^2 + z_2^2)^{1/2}}{2a_0} \right]$$

Hence, in typical coordinates,

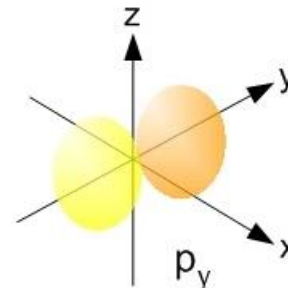
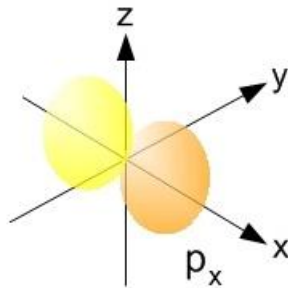
$$\hat{C}_{4(z)}\psi_{2p_x} = y \exp \left[-\frac{(x^2 + y^2 + z^2)^{1/2}}{2a_0} \right] = \psi_{2p_y}$$

Example 13.14

The normalized $2p_x$ wave function for the electron in a hydrogen atom is

$$\psi_{2p_x}(x, y, z) = x \exp \left[-\frac{(x^2 + y^2 + z^2)^{1/2}}{2a_0} \right],$$

where a_0 is the Bohr radius. Find $\hat{C}_{4(z)}\psi_{2p_x}$.



(Image: https://commons.wikimedia.org/wiki/File:Single_electron_orbitals.jpg)

Eigenfunctions of Symmetry Operators

- Eigenvalue = +1

the operator leaves a function unchanged

- Eigenvalue = -1

the operator changes the sign of the function's value at all points

Exercise 13.14

(a) Find $\hat{i}\psi_{2px}$; (b) show that ψ_{2px} is an eigenfunction of the inversion operator; and (c) find its eigenvalue.

Since $\hat{i}(x_1, y_1, z_1) = (x_2, y_2, z_2) = (-x_1, -y_1, -z_1)$, convert

$$x_2 \leftarrow -x_1, \quad y_2 \leftarrow -y_1, \quad z_2 \leftarrow -z_1;$$

$$\psi_{2px}(x_1, y_1, z_1) = x_1 \exp \left[-\frac{(x_1^2 + y_1^2 + z_1^2)^{1/2}}{2a_0} \right]$$

$$(\hat{i}\psi_{2px})(x_2, y_2, z_2) = -x_2 \exp \left[-\frac{\{(-x_2)^2 + (-y_2)^2 + (-z_2)^2\}^{1/2}}{2a_0} \right]$$

$$\hat{i}\psi_{2px} = -x \exp \left[-\frac{(x^2 + y^2 + z^2)^{1/2}}{2a_0} \right]$$

Exercise 13.14

(a) Find $\hat{i}\psi_{2px}$; (b) show that ψ_{2px} is an eigenfunction of the inversion operator; and (c) find its eigenvalue.

$$\hat{i}\psi_{2px} = -x \exp \left[-\frac{(x^2 + y^2 + z^2)^{1/2}}{2a_0} \right]$$

which reads that

$$\hat{i}\psi_{2px} = -\psi_{2px}$$

This is the eigenvalue equation, and it shows that ψ_{2px} is an eigenfunction of the inversion operator. The eigenvalue is -1 .